`scripts/verify-codegraph.sh` — user guide

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Overview

`scripts/verify-codegraph.sh` is the anti-bluff verification suite runner for the **codegraph** code-intelligence integration (see `docs/CODEGRAPH.md` and constitution §11.4.78). Per the Local-Only CI/CD constitutional constraint it **is** the codegraph quality gate — there is no hosted-CI equivalent.

It runs the six `tests/codegraph/test_*.sh` layers in order, tees per-layer evidence to `.lava-ci-evidence/codegraph/<UTC-timestamp>/`, writes a `summary.md`, and exits non-zero if any layer FAILs.

Prerequisites

- `codegraph` installed and on `PATH` (`npm install -g @colbymchenry/codegraph`).
- `.codegraph/codegraph.db` present — run `codegraph index` first if absent.
- `python3` (used to drive the MCP JSON-RPC layers).
- For the full run (layer 05): the 5 CLI agents (`claude`, `opencode`, `qwen`, `kimi`, `crush`) installed; un-authenticated agents are reported as documented SKIP gaps, never as failures.

Usage

```
scripts/verify-codegraph.sh              # full run — all 6 layers, incl. agent E2E
scripts/verify-codegraph.sh --quick      # layers 01-04 + 06 only (skip slow LLM layer 05)
```

Environment overrides:

- `CODEGRAPH_EVIDENCE_DIR` — override the evidence output directory.
- `CODEGRAPH_AGENT_TIMEOUT` — per-agent timeout in seconds for layer 05 (default 240).

The six layers

Layer	File	Proves
01	`test_01_index_reality.sh`	codegraph indexed the real Lava codebase; secrets/submodules excluded
02	`test_02_query_correctness.sh`	`codegraph query` resolves real symbols to real `file:line`
03	`test_03_mcp_protocol.sh`	the MCP server returns real data over real stdio JSON-RPC
04	`test_04_agent_connectivity.sh`	all 5 CLI agents register/connect to codegraph
05	`test_05_agent_e2e.sh`	each agent provably *uses* codegraph in a real LLM turn (unforgeable node-count challenge)
06	`test_06_falsifiability.sh`	the suite FAILs when codegraph is deliberately broken, PASSes when restored

Edge cases

- **`--quick`** skips the slow LLM-driven layer 05 — use it for fast local checks; the full run is required before relying on the integration.
- **Documented gaps (SKIP):** layer 05 marks an agent SKIP when it cannot be run in the environment (missing credentials, exhausted quota, an agent-environment incompatibility). A SKIP is a documented gap, never a faked PASS (§6.J / §6.L); it does not fail the suite, but is reported prominently.
- **Pre-flight failure (exit 2):** if `codegraph` is not installed or `.codegraph/codegraph.db` is missing, the runner aborts before running any layer.
- **Falsifiability rehearsal** (layer 06) moves `.codegraph/codegraph.db` aside and restores it; a `trap` restores the DB even if interrupted.

Internal behaviour

The runner sources nothing; each `tests/codegraph/test_*.sh` sources `tests/codegraph/lib.sh` for shared assertions and the MCP JSON-RPC drivers. Each layer exits 0 (all assertions pass) or 1 (any fail). The runner aggregates PASS/FAIL/SKIP counts from the per-layer logs and writes `summary.md`. Exit codes: `0` suite passed, `1` a layer FAILed, `2` pre-flight failed.

Related scripts

- `tests/codegraph/lib.sh` — shared library (assertions, MCP drivers).
- `tests/codegraph/test_0[1-6]_*.sh` — the six verification layers.
- `docs/CODEGRAPH.md` — the codegraph integration reference.

Last verified

2026-05-20 — full run: 45 pass · 0 fail · 4 documented SKIP gaps.